

Complex ResNet Aided DoA Estimation for Near-Field MIMO Systems

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Abstract—The near-field effect of short-range multiple-input multiple-output (MIMO) systems imposes many challenges on direction-of-arrival (DoA) estimation. Most conventional scenarios assume that the far-field planar wavefronts hold. In this paper, we investigate the DoA estimation problem in short-range MIMO communications, where the effect of near-field spherical wave is non-negligible. By converting it into a regression task, a novel DoA estimation framework based on complex-valued deep learning (CVDL) is proposed for the near-field region in short-range MIMO communication systems. Under the assumption of a spherical wave model, the array steering vector is determined by both the distance and the direction. However, solving this regression task containing a massive number of variables is challenging, since datasets need to capture numerous complicated feature representations. To overcome this, a virtual covariance matrix (VCM) based on received signals is constructed, and thus such features extracted from the VCM can deal with the complicated coupling relationship between the direction and the distance. Although the emergence of wireless big data driven by future communication networks promotes deep learning-based wireless signal processing, the learning algorithms of complex-valued signals are still ongoing. This paper proposes a one-dimensional (1-D) residual network that can directly tackle complex-valued features due to the inherent 1-D structure of signal subspace vectors. In addition, we put forth a cropped VCM based policy which can be applied to different antenna sizes. The proposed method is able to fully exploit the complex-valued information. Our simulation results demonstrate the superiority of the proposed CVDL approach over the baseline schemes in terms of the accuracy of DoA estimation.

I. INTRODUCTION

With the advances of key enabling technologies in the fifth generation (5G) mobile communications, such as millimeter wave (mmWave), small cell, and massive multiple-input multiple-output (MIMO), the direction-of-arrival (DoA) estimation problem [1] in the context of massive MIMO systems has attracted substantial research in recent years [2]. In past decades, many DoA estimation methods have been proposed, varying from subspace-based techniques [3]–[5], maximum-likelihood estimators [6], compressed sensing methods [7], to hybrid methods [8]–[10]. Efficient and accurate

DoA estimation algorithms are crucial to massive MIMO systems [11]–[13]. Specifically, accurate DoA information is shown to be critical for beamforming design in massive MIMO systems [14].

Most of DoA estimation methods mentioned above assume far-field planar wavefront. However, this assumption may be invalid for future mobile systems. As a promising enabling technique for 5G-and-beyond mobile networks, mmWave massive MIMO is expected to support high data-rate communications. However, the short wavelength and large antenna aperture induce challenges in the modeling of propagation patterns. When the Rayleigh distance is considered, the last-order scatterers fall within the near-field region of the receive antenna, where spherical wave propagation is applicable [15]. On the other hand, the near-field DoA estimation appears to be vitally important in many other applications, including vehicle-to-everything (V2X) communications [16], [17], virtual reality, smart home, and automated driving. Particularly, the DoA estimation of signals from landmarks is the core technology of vehicle positioning [18], which is an important branch in the Internet of Vehicles (IoV).

A. Related Work and Motivation

In near-field DoA estimation, the joint estimation of the distance and the angle is inevitable, which extends the search space and increases the search overhead. To overcome this issue, a subspace method without multidimensional search is proposed in [19]. These subspace methods require many snapshots and high signal-to-noise ratio (SNR) to improve the spatial resolution. Furthermore, a maximum-likelihood-based approach [20] by iterative estimation is proposed for the case of a small number of snapshots and low SNR. However, as the parameter dimension increases, the iterative methods inevitably face the predicament of slow processing speed.

Meanwhile, wireless networks are generating a massive volume of data, from which we can extract useful information using the state-of-the-art machine learning (ML) techniques [21]–[24]. There have been some efforts exploring the applications of ML in DoA estimation. In [25], the near-field DoA estimation is modeled as a classification problem based on the support vector machine (SVM) approach. The authors of [26] extend the application of deep neural networks (DNNs) to a general far-field acoustic DoA estimation problem, which is modeled as a classification problem and not suitable for near-field estimation. Although these ML-based classification methods are interesting, their accuracy depends on the quantization

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resolution of angles. In [27], a support vector regression (SVR) method is proposed for estimating the DoAs of near field sources. Popular deep learning (DL) methods are also used in far-field DoA estimation such as in the field of acoustic source direction finding [28]. However, such models of acoustic DoA estimation cannot be directly applied to signal processing in wireless systems. In addition, a real-valued DNN (RVNN) for estimating DoA in hybrid massive MIMO channels is designed in [29]. Traditionally, complex-valued signals are split into the real and imaginary parts, which are real values and fed into the network [26], [29]–[31]. Nonetheless, this treatment may fail to capture the correlation between the real part and the imaginary part, thus incurring the phase information loss.

Most of the existing DL methods focus on the real domain. However, many wireless communications problems to be solved are in the complex domain. Merging the conception of complex-valued neural network (CVNN) with wireless signal processing may provide a further thrust to enable smart radio. The complex-valued deep learning (CVDL) methods refer to the DNNs that can perform complex arithmetic, and the weight parameters of CVNNs are complex-valued [32], [33]. The authors of [34]–[36] explored the applications of CVDL in image and speech processing. These network architectures are all designed to account for image classification, music transcription or speech spectrum prediction. Against the above backdrop, our motivations can be explained from the following aspects:

- Since the conventional near-field methods require high SNR and snapshots or online iterative estimation, it is unlikely to estimate DoA in real time. Moreover, there is a very limited study about designing DL methods for near-field DoA estimation. As DL is a powerful tool to solve intractable nonlinear problems, it would be interesting to improve the performance robustness if training an end-to-end network to solve the near-field DoA estimation.
- Both spatial spectrum search and classification-based ML methods cannot meet the precision requirements when using coarse grids. Besides, RVNNs may destroy the structure of complex data via splitting real and imaginary parts. Thus, a CVNN-based regression model is expected to address these issues.
- When near-field signals are concerned, the CVDL-based DoA estimation is not straightforward due to the coupling between the DoA and range parameters. How to design input features, activation functions and loss metrics motivates us to pursue a general CVDL-based DoA estimation framework.

B. Main Contributions

In this paper, a novel CVDL-aided DoA estimation approach is proposed for near-field MIMO systems, where the DoA estimation is viewed as an end-to-end regression task rather than a classification problem. First, to decouple the distance and the direction, we design a new feature representation based on the reconstructed virtual covariance matrix (VCM). This representation can eliminate the learning obstacles in DoA estimation. Then, a complex-valued residual network (ResNet)

is proposed to learn complex features of the input data. Compared with the RVNN, the proposed network circumvents the trivial split and splice between real and imaginary parts. To the best of our knowledge, it is the first time that the phase mapping activation function is developed for the CVNN. How to introduce CVDL into massive MIMO systems has not been well addressed before.

The main contributions of this paper are summarized as follows:

- From an ML perspective, we investigate the near-field DoA estimation problem, which is considered as a phase feature extraction process of input data. Without the need for tedious grid-based classification, we train an end-to-end deep network for angle estimation. This network is capable of processing complex-valued tasks directly.
- New features, based on the reconstructed VCM instead of the whole or half covariance matrix, are designed as the inputs of the CVNN. Therefore, the computational complexity of our deep networks is greatly decreased. Moreover, the joint two-dimensional (2-D) parameter estimation problem in the near-field is transformed into a one-dimensional (1-D) DoA estimation problem, where the designed input features are able to eliminate the complicated coupling relationships between the distance and the direction.
- A 1-D convolutional complex-valued ResNet is proposed to process 1-D complex features. We design a residual block architecture with a shortcut connection to ensure the training stability. To map the real-valued angles from the complex-valued output, we devise a phase mapping activation function specifically for the DoA estimation task. Besides, we also carefully select different complex-valued activation functions in the corresponding neuron layers according to the distribution of the dataset.
- For our specific regression task, we compare the training results of different loss metrics and determine the suitable loss function through comparison. As a result, the prediction precision is enhanced.
- To realize a general CVNN model which can be applicable to different antenna sizes, we put forth a cropped VCM based scheme. This scheme can be deployed for arbitrary antenna size without the need for redesigning new networks. Moreover, it can reduce the computational cost since the inputs are cropped to a fixed shape.

The remainder of this paper is organized as follows. In Section II, we establish the received signal model for the near-field region. In Section III, the VCM reconstruction method is proposed to remove the corruption of range parameters on the signal subspace; and the feature preprocessing scheme is proposed based on the VCM. In Section IV, we present the CVDL-aided DoA estimation framework and introduce the complex ResNet with specific model settings. Our simulation results are provided in Section V, which demonstrates that the CVNNs outperform the baseline methods. Our conclusions are offered in Section VI.

Notations: Lower-case boldface symbols indicate column vectors, and upper-case boldface symbols refer to matrices;

$$\left| [\mathbf{R}]_{p,p+t} - [\bar{\mathbf{R}}]_{p,p+t} \right|^2 \approx \sum_{m=1}^M \left| [\mathbf{R}_s]_{m,m} \right|^2 \left[\left((t + 2\delta_p) \frac{d \sin \theta_m}{r_m} \right)^2 + \left(\kappa_{p,m} \kappa_{q,m} t (t + 2\delta_p) \frac{\pi d^2}{\lambda r_m} \cos^2 \theta_m \right)^2 \right]. \quad (22)$$

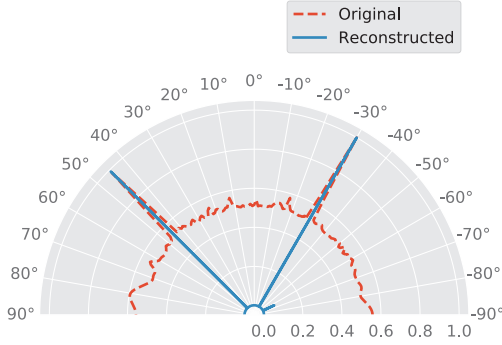


Fig. 3. Beam patterns of original and reconstructed covariance matrices with two signals, which are propagated from -30° and 45° .

matrix $\hat{\mathbf{E}}_s$, i.e.,

$$\text{span}[\xi_1, \dots, \xi_M] = \text{span}[\mathbf{a}_1(\theta_1), \dots, \mathbf{a}_M(\theta_M)], \quad (25)$$

where $\text{span}[\cdot]$ stands for the span of a set of vectors.

IV. PROPOSED DEEP COMPLEX NETWORK ARCHITECTURE

A. Deep Complex Network Architecture

In signal processing, for the convenience of signal representation, we usually deal with complex-valued signals that contain both the amplitude and phase information. Inspired by [34] and [35], we design a deep complex network for meaningful information extraction, which is used to carry out the DoA estimation.

The proposed complex-valued network is able to simulate the complex arithmetic operations in an internal mechanism. Let c denote a complex number in the form of $c = u + jv$ with $u, v \in \mathbb{R}$. Assume that the input of the l -th layer of the network is a complex vector $\mathbf{c}^l = \mathbf{u}^l + j\mathbf{v}^l$, where $\mathbf{u}^l$ and $\mathbf{v}^l$ are real vectors. In the l -th layer of the proposed CVNN model, the complex weight matrix $\mathbf{W}^l$ and complex bias vector $\mathbf{b}^l$ are expressed as

$$\mathbf{W}^l = \mathbf{W}_R^l + j\mathbf{W}_I^l, \quad \mathbf{b}^l = \mathbf{b}_R^l + j\mathbf{b}_I^l, \quad (26)$$

where $\mathbf{W}_R^l$ and $\mathbf{W}_I^l$ correspond to the real and imaginary parts of $\mathbf{W}^l$, respectively; $\mathbf{b}_R^l$ and $\mathbf{b}_I^l$ denote the real and imaginary parts of $\mathbf{b}^l$, respectively.

Empirically, the depth of the network plays an important role in classification or regression tasks from a training point of view. However, deeper networks may result in more intractable issues, such as the gradient exploding or vanishing. Inspired by the ResNet proposed in [43], we develop a residual block architecture consisting of several blocks, where a shortcut connection is added to realize the linear superposition of the input itself and its nonlinear transformation in each block. The specific components of each block are shown in Fig. 4.

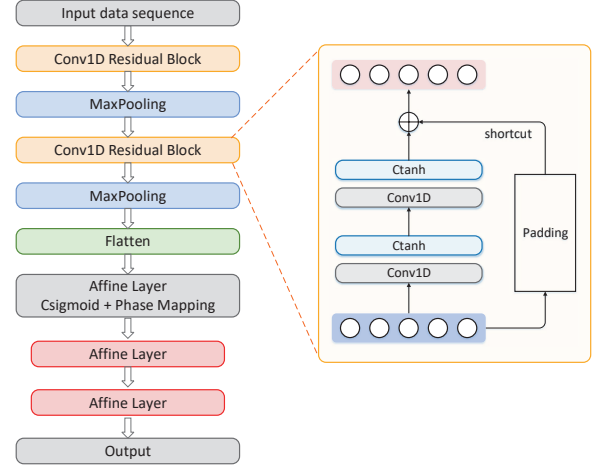


Fig. 4. The deep complex ResNet model. The network is shown on the left side, and the right side represents the Conv1D residual block diagram.

As shown in Fig. 4, the architecture of the proposed complex ResNet is composed of two alternations of convolutional and pooling layers, a flatten layer, and three affine layers. In our proposed deep complex network, two 1-D convolution (Conv1D) layers are contained in a convolutional residual block. We mainly employ 2 residual blocks to learn the phase-related features of the input complex-valued data. The residual block is followed by a pooling layer, which extracts the critical features. The results after pooling are flattened into a column and fed into a complex affine layer, whose outputs are mapped to real-valued phases by specially devised activation functions. Finally, since we define the near-field DoA estimation as a regression problem, several real affine layers must be connected to the end of network to predict the continuous phase variable. The details of each type of layer representation are listed in the following.

1) Affine layer: Affine means that the neurons between adjacent layers are fully connected. The complex-valued arithmetic process is expressed as

$$\begin{bmatrix} \Re(\mathbf{W}^l \mathbf{c}^l) \\ \Im(\mathbf{W}^l \mathbf{c}^l) \end{bmatrix} = \begin{bmatrix} \mathbf{W}_R^l & -\mathbf{W}_I^l \\ \mathbf{W}_I^l & \mathbf{W}_R^l \end{bmatrix} \begin{bmatrix} \mathbf{u}^l \\ \mathbf{v}^l \end{bmatrix}. \quad (27)$$

The signal flow diagram of the complex-valued affine layer is shown in Fig. 5.

2) Convolution layer: The convolution operation of the CVNN, analogous to affine layer, is expressed as

$$\begin{aligned} \mathbf{W}^l \otimes \mathbf{c}^l = & (\mathbf{W}_R^l \otimes \mathbf{u}^l - \mathbf{W}_I^l \otimes \mathbf{v}^l) \\ & + j(\mathbf{W}_I^l \otimes \mathbf{u}^l + \mathbf{W}_R^l \otimes \mathbf{v}^l). \end{aligned} \quad (28)$$

Additionally, the Conv1D filters are employed in 1-D input features analysis. If the size of the input matrix is $L_i \times F_i$, where L_i is the length of the input vector and F_i is the number

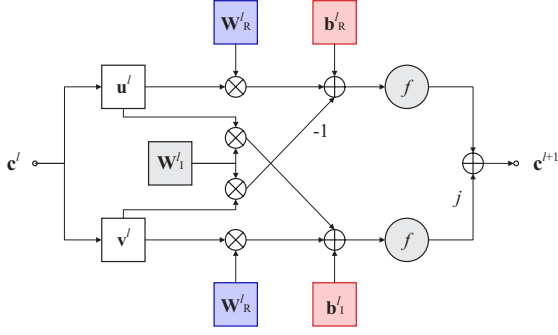


Fig. 5. The schematic model of the affine layer of the proposed CVNN.

of the feature channels, then the required filter has a dimension of $L_c \times F_i \times F_c \times S_c$, where S_c is the stride used to describe the step size of the convolution operation and L_c is the length of the filter. In addition, F_c represents the number of filters, and it determines the number of output channels. A filling-up convolution method, known as “SAME” padding, is conducted to avoid losing information from the original data. Then, the dimension of the output after Conv1D will be $\lceil L_i/S_c \rceil \times F_c = L_{co} \times F_c$ for “SAME” padding. After applying the 1-D pooling layer with the $L_p \times S_p$ kernel, the final size of the output becomes $\lceil L_{co}/S_p \rceil \times F_c$, where L_p denotes the length of the pooling kernel and S_p is the stride of the pooling kernel.

3) Pooling layer: In convolutional neural networks (CNNs), the pooling layer is often placed behind the convolution layer to incorporate the features from small neighborhoods. As a result, the key features for prediction can be obtained, while sparse and valid information can be retained. In this paper, the max-pooling layers are applied to the real and imaginary parts, respectively.

B. Activation Function and Backpropagation

For CVNNs, the activation function is the most critical part, since it directly determines the nonlinear fitting ability and the convergence of the back propagation (BP) algorithms [30]. Generally speaking, complex activation functions should be analytic and bounded. In other words, the functions should satisfy the Cauchy-Riemann (C-R) equation. In the complex analysis, Liouville’s theorem states that complex functions that are both analytic and bounded must be constant functions [36]. However, in existing literature, many researchers prefer bounded functions to analytic functions. Simultaneously, due to the fact that the phase information is determined by both the real and the imaginary parts, we consider an activation function that affects the magnitude to indirectly achieve the phase learning. To this end, a split complex function is employed in convolution layers, and it is given by

$$\text{Ctanh}(c) = \tanh(u) + j \tanh(v). \quad (29)$$

Although the activation function $f(c) = \text{Ctanh}(c)$ is non-analytic, it is bounded in the complex domain. The boundedness of the activation function eliminates the appearance of singular points. Considering that our training targets have values in the interval $(-\pi/2, \pi/2)$, it will cause a large angle

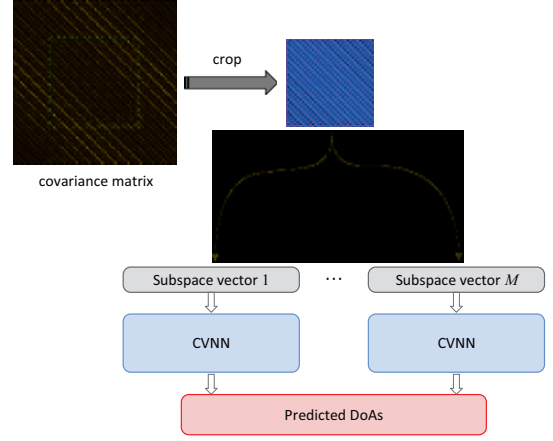


Fig. 6. The CVDL based DoA estimation framework.

error in degrees even when the small training loss is reached. Rescaling from the target values in radians does not work. Note that the mean of the target data is zero. To make the model focus more on small differences, we adopt the tanh function in affine layers since it has the highest slope at the zero point. Such a design can cause the gradient boosting with small prediction differences.

We also devise a special activation function for a mapping from the complex domain to the real domain, thus facilitating the measurement of prediction errors. To be specific, after the complex affine layer we define the following activation functions:

$$\text{Csigmoid}(c) = \text{sigmoid}(u) + j \text{sigmoid}(v), \quad (30)$$

$$\rho = \arctan \frac{\Im(c)}{\Re(c)} = \arctan \left(\frac{c - c^*}{c + c^*} \right), \quad (31)$$

where ρ is the real-valued phase corresponding to the output of complex affine layer. To avoid the invalid division in (31), we employ the sigmoid function in (30) to keep the real part of the output greater than 0.

In CVNNs, the derivable function $\mathcal{L}$ regarding the real and imaginary parts of parameters can also be applied to BP, respectively [44]. The derivative processing for the real and imaginary parts is written as

$$\frac{\partial \mathcal{L}}{\partial \mathbf{c}} = \Re(\nabla \mathcal{L}(\mathbf{c})) + j \Im(\nabla \mathcal{L}(\mathbf{c})), \quad (32)$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}} = \Re(\nabla \mathcal{L}(\mathbf{c})) \left(\frac{\partial \mathbf{u}}{\partial \mathbf{W}_R} + j \frac{\partial \mathbf{u}}{\partial \mathbf{W}_I} \right) + \Im(\nabla \mathcal{L}(\mathbf{c})) \left(\frac{\partial \mathbf{v}}{\partial \mathbf{W}_R} + j \frac{\partial \mathbf{v}}{\partial \mathbf{W}_I} \right). \quad (33)$$

Accordingly, the network parameters $\mathbf{W}_R^l$ and $\mathbf{W}_I^l$ are iteratively updated by means of the stochastic gradient descent (SGD) algorithm [30].

C. Training Strategy

We intend to propose a strategy which can be applied to different numbers of antennas at the receiver array, instead of training special DNNs with different input dimensions corresponding to various numbers of antenna elements. Different

from the existing methods, we design new input features for a general deep network in our DoA estimation framework. The training strategy can be seen from Fig. 6. Above all, the VCM derived from any N -element antenna is cropped to a fixed size so as to retain the key features. The cropped VCM is constructed as

$$\begin{aligned} \underline{\mathbf{R}} &= \left[\hat{\underline{\mathbf{R}}} \right]_{\frac{N-N_{\text{in}}}{2}+1:N-\frac{N-N_{\text{in}}}{2}+1, \frac{N-N_{\text{in}}}{2}+1:N-\frac{N-N_{\text{in}}}{2}+1} \\ &= \underline{\Xi}_s \underline{\Lambda}_s \underline{\Xi}_s^H + \underline{\Xi}_z \underline{\Lambda}_z \underline{\Xi}_z^H, \end{aligned} \quad (34)$$

where $N_{\text{in}} \times N_{\text{in}} (N_{\text{in}} < N)$ is the cropped block size of the original VCM. Given the signal subspace vectors $\underline{\Xi}_s = [\underline{\xi}_1, \dots, \underline{\xi}_M]$, the input features are constructed as

$$\mathbf{c}_{\text{input}} = \underline{\xi}_m \in \mathbb{C}^{N_{\text{in}}}, \quad m = 1, 2, \dots, M. \quad (35)$$

Finally, independent subspace vectors are adopted as training samples to perform the learning task.

Algorithm 1 The proposed CVDL-aided DoA estimation method

Input:

The received signals $\mathbf{y}(k)$, $k = 1, 2, \dots, K$;
The number of near-field sources M ;

Output:

The estimates of DoA, $\hat{\theta}$;

- 1: Calculate the covariance matrix $\mathbf{R}$ based on the samples $\mathbf{y}(k)$ according to (10);
- 2: Reconstruct the VCM $\hat{\mathbf{R}}$ according to

$$[\hat{\mathbf{R}}]_{p,p+t} = \{[\mathbf{R}]_{\delta_l(t), \delta_l(-t)} + [\mathbf{R}]_{\chi_r(t), \chi_r(-t)}\} / 2;$$

- 3: Crop the VCM $\hat{\mathbf{R}}$ obtained by (23) to acquire the compressed signal subspace $\underline{\Xi}_s$ based on (34);
- 4: Devise the input feature representations $\mathbf{c}_{\text{input}}$ based on (35) to build the training samples;
- 5: Feed the data into the proposed CVNN depicted in Fig. 4, and perform forward propagation to obtain the network output $\hat{\theta}$;
- 6: **return** $\hat{\theta}$;

In this paper, more attention is paid to angle prediction in regression tasks. Both mean squared error (MSE) and mean absolute error (MAE) are popularly used for regression loss metrics [45]. Although the MSE loss is easy to compute the gradient, large errors have relatively greater influence on the MSE than the small errors due to the square. Compared with the MSE, the MAE loss is more robust to outliers since it does not use square. Moreover, the MAE is more useful if we are concerned about small errors such as subtle radian errors. Therefore, a typical MAE loss function is defined as

$$\mathcal{L}(\hat{\theta}, \theta) = \frac{1}{BM} \sum_{b=1}^B \|\hat{\theta}_b - \theta_b\|_1, \quad (36)$$

where B is the total number of data samples. The true and estimated DoAs for the b -th sample are denoted by θ_b and $\hat{\theta}_b$, respectively.

For the purpose of extending to multi-source DoA estimation, we employ a training strategy that only uses the signal subspace vector as training samples. As such, the subspace vectors associated with multi-DoA are fed into the network

Table I: CVNN Hyperparameters

Layer Name		Configuration
Input		$N_{\text{in}} \times 1$
Residual Block 1	Conv1D $\times 2$	$3 \times 1 \times 8 \times 1$
	Conv1D $\times 2$	$3 \times 8 \times 8 \times 1$
MaxPooling		2×2
Residual Block 2	Conv1D $\times 2$	$3 \times 8 \times 4 \times 1$
	Conv1D $\times 2$	$3 \times 4 \times 4 \times 1$
MaxPooling		2×2
Flatten		N_{flat}
Affine (Csigmoid + Phase Mapping)		20
Affine		10
Affine		10
Output		1

Table II: RVNN (TDNN [30]) Architecture

Layer Name	Layer Specification
input layer	input dim = N'_{in}
hidden layer 1	context size = 5, filter num = 8
hidden layer 2	context size = 5, filter num = 8
hidden layer 3	context size = 5, filter num = 4
hidden layer 4	context size = 5, filter num = 2
hidden layer 5	context size = 5, filter num = 1
dense layer	output dim = 10
dense layer	output dim = 10
output layer	output dim = 1

separately to train and predict. The proposed CVDL-aided DoA estimation method is summarised in Algorithm 1.

D. Complexity Analysis

We use the total number of floating-point operations (FLOPs) to measure the computational complexity of the proposed network. The computational complexities of both the convolutional and dense layer operations are 4 times more expensive than their real counterparts but with the same order [35]. According to [46], in real-valued networks, the FLOPs of a conv1D layer is given by $2L_i(F_i L_c^2 + 1)F_c$ and the FLOPs of a dense layer is given by $(2I - 1)O$, where I and O denote the input and output dimensions, respectively.

Regarding the deep complex model, the hyperparameters of the CVNN are detailed in Table I. These parameters are listed according to the order of the layers depicted in Fig. 4, and the network configuration $L_c \times F_i \times F_c \times S_c$ corresponds to the numerical expression of the Conv1D kernels in the third column of Table I. For the training and test sets, each sample of the dataset contains N_{in} entries taken from the signal subspace $\underline{\Xi}_s$ generated by using (34). The number of neurons in the flatten layer is denoted as N_{flat} .

The time delay neural network (TDNN) architecture proposed in [30] has been taken as the RVNN-based baseline scheme for regression prediction, and its structure and parameter settings are shown in Table II. Since the RVNN cannot directly deal with the complex-valued features, according to

[30], the real and imaginary parts of (35) are concatenated as the input tensor, i.e.,

$$\mathbf{u}_{\text{input}} = \begin{bmatrix} \Re(\mathbf{c}_{\text{input}}) \\ \Im(\mathbf{c}_{\text{input}}) \end{bmatrix} \in \mathbb{R}^{N'_{\text{in}}}, \quad (37)$$

where $N'_{\text{in}} = 2 \times N_{\text{in}}$ indicates that the input size of the RVNN is twice as the input size of the CVNN.

For the case of $N_{\text{in}} = 33$, the total number of FLOPs of the proposed CVNN is about 0.24 million. In this case, we have $N'_{\text{in}} = 66$ for the real TDNN and its total number of FLOPs is about 0.34 million. Note that the proposed CVNN can keep operations within complex structures while conventional RVNNs achieve brute-force fitting by increasing network depth and dimensions.

Table III: Parameters of the Near-Field Source Dataset

Parameters	Value and Range	
	Training Set	Test Set
Distance (in λ)	(200, 1800) step: 25	(200, 1800) step: 25
Direction ($^\circ$)	(-90, 90) step: 0.01	(-90, 90) step: 0.1
Snapshots	100	100
SNR (dB)	10	10

V. SIMULATION RESULTS AND DISCUSSIONS

A. Simulation Setup

In this section, the CVDL-aided DoA estimation approach is evaluated by numerous simulations. All parameter settings are in accordance with the near-field region. The implementation of our deep network is based on TensorFlow [47]. The estimation results are compared with the MUSIC method in [19], the SVR method in [27] and the real TDNN scheme in [30]. The root mean square error (RMSE) is used to evaluate our proposed model.

To elaborate a little further, the transmitted signals with carrier frequency of $f_c = 28$ GHz are used. The corresponding wavelength is $\lambda = 0.0107$ m. We consider a ULA with $N = 65$ array elements, thus the position of the reference array element is $n_c = 33$. We set the array spacing as $d = \lambda/2$. Then, the array aperture is $D = 32\lambda$, and the Fresnel region is between 112λ and 2048λ [41].

For the dataset, the received signals are collected by using (9), and the input datasets are generated according to the scheme proposed in Section III-B. In attempting to capture the impact of multiple factors, the dataset should cover a variety of conditions, such as snapshots, directions and distance from the reference array element, which are summarized in Table III. Note that we train the CVNN offline by employing the raw data with additive noise as the training samples, where the SNR and the number of snapshots are fixed at 10 dB and 100, respectively. Then we will test the performance of online deployment under various SNRs and snapshots. Based on our CVNN framework, the learning model of multi-source scenarios can be trained via adding subspace vector samples associated with multi-DoA. As an initial work and for the sake of simplicity, we take $M = 1$ as an example.

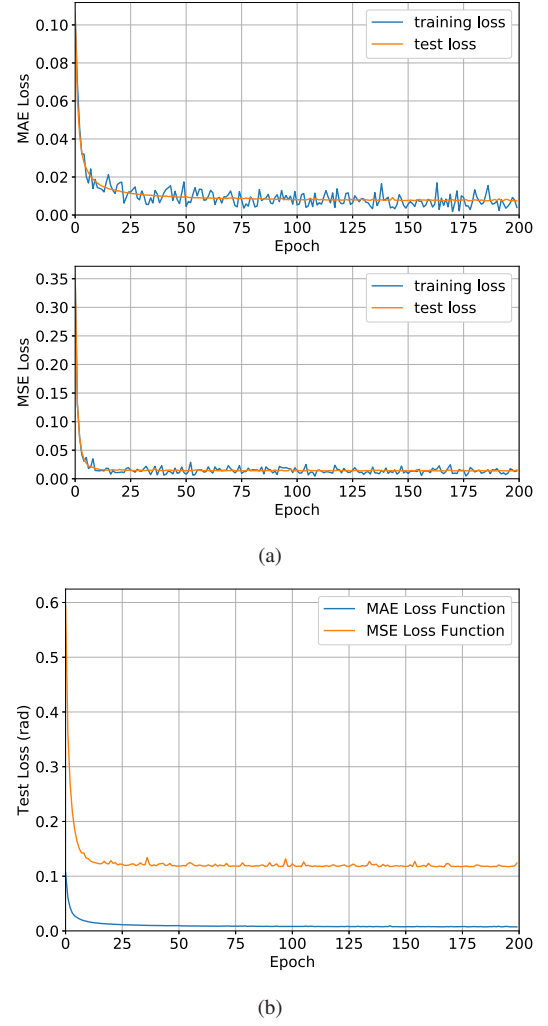


Fig. 7. Model training results for different loss functions. (a) Learning curve of 200 epochs in the offline training phase. (b) Comparing test errors trained with different loss metrics.

During the training process, the SGD method is employed to speed up the training, and the adaptive moment estimation (Adam) algorithm [48] is used to update the parameters. To investigate the loss functions that are appropriate for our regression predictive modeling problems, Fig. 7 analyzes the effects of different loss metrics (i.e., MAE and MSE) on learning behavior and test performance. Fig. 7(a) shows that both models converge to a steady state and the MSE loss are declining faster than the MAE loss. This is because the gradient of the MSE loss is high for larger loss values at initial epochs. The spikes appearing in the learning curve are the unavoidable consequences of the mini-batch GD methods. We plot Fig. 7(b) to compare the test errors (in rad) trained with the MAE and MSE loss metrics. It is seen that the MAE-based test error can reach far below 0.05 after convergence, but the MSE-based test error is above 0.1. This validates the effectiveness and suitability of the MAE loss for our regression predictive task.

In addition, we also study the impacts of residual blocks on the whole network performance. We train different networks